JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING (AFFILIATED TO JNTU-GV)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

★ OFFICIAL ACADEMIC SYLLABUS & BLUEPRINT ★

B.Tech ECE R23 Electronic Devices and Circuits (EDC) Syllabus & Question Bank

Course Code:	R23-ECE-1102	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit I: Semiconductor Physics & PN Junction Diode

Intrinsic and extrinsic semiconductors, Carrier transport (drift and diffusion), PN junction diode under forward and reverse bias, Diode current equation, Transition and diffusion capacitance, Zener diode and voltage regulation.

Unit II: Rectifiers & Filters

Half-wave, full-wave center-tapped and bridge rectifiers, Ripple factor, Efficiency, TUF, Harmonic components, Inductor filter, Capacitor filter, L-section and Pi-filters.

Unit III: Bipolar Junction Transistors (BJT)

Transistor construction, CE, CB, and CC configurations, Input and output characteristics, Transistor as a switch and amplifier, Early effect, Punch-through.

Unit IV: Transistor Biasing & Thermal Stabilization

Need for biasing, Operating point (Q-point), Stability factors S, S', S'', Fixed bias, Collector-to-base bias, Self-bias (Voltage divider bias), Thermal runaway.

Unit V: Field Effect Transistors (JFET & MOSFET)

Construction and operation of N-channel JFET, Drain and transfer characteristics, Pinch-off voltage, MOSFET (Depletion and Enhancement types), Comparison of BJT and FET.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Internal Marks	30 Marks
External Examination	70 Marks
Reference Book	Electronic Devices and Circuit Theory by Boylestad / Millman & Halkias

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM